REVIEWER'S COMPANION TO *SMALL MAXIMAL K-CONSECUTIVE SUMS ON A CIRCLE AND THE CLOCK-FACE PROBLEM*

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Abstract. This companion answers three review points. Section 1 traces every hypothesis of every result in the note to the step that uses it, and lists the only outside results the proofs need. Section 2 proves the upper-bound construction (Theorem 2) again with every index written out, including why the order of the value blocks never matters. Section 3 goes through the boundary cases. Section 4 says what the computer checks do and do not establish. Numbering of lemmas and theorems refers to the note (version of November 2026).

1. Hypotheses and logical dependencies

Throughout, positions are taken modulo n, $[n] = \{1, \dots, n\}$, and $s_i = \pi_i + \cdots + \pi_{i+k-1}$. No step of any proof uses a computer. The only results imported from the literature are two upper bounds of Anstee, Ferguson and Griggs, quoted from Higashitani and Kurimoto, Theorem 2.4:

(AFG4) $\mathsf{disc}(n, 3) \le 2$ for $n \ge 6$ (AFG, Theorem 4; HK, Theorem 2.4(iii));
(AFG7) $\mathsf{disc}(mk, k) \le 2$ for odd k (AFG, Theorem 7; HK, Theorem 2.4(iv)).

Both are used only through $\mathsf{msum} \le \mathsf{disc}$, which holds by definition. Everything else cited in the note (the values $\mathsf{msum}(15, 3)$, $\mathsf{msum}(21, 3)$, $\mathsf{msum}(35, 7)$, HK's Theorem 1.1 and Lemma 5.3, the history) is context and is not used in any proof.

Result	Hypotheses	Uses	Where each hypothesis is used
Lemma 5	$0 < k < n$; $s_i \le L$, $L \in \mathbb{Z}$	definitions only	(i) $0 < k < n$: positions i, $i + k$ differ, so $\pi_{i+k} \ne \pi_i$. (iii) $L \in \mathbb{Z}$: the σ_i are integers, so a positive pair contributes ≥ 1 to M.
Chains, $D \ge 2g$	$0 < k < n$	definitions	$g = \gcd(n, k) \le k < n$ gives chain length $\ell = n/g \ge 2$, so every chain has a peak and a valley.
Lemma 6	$0 < k < n$	chain facts	distinct values on a chain give $P^* - V^* \ge \ell - 1$; holds for every $\ell \ge 2$ (for $\ell = 2$ the chain is one peak and one valley).
Prop. 7	as Lemma 5	Lemma 5(ii), Lemma 6	none beyond those.

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Result	Hypotheses	Uses	Where each hypothesis is used
Lemma 8	k odd; $\sigma_i \geq 0$	Lemma 5(i)	k odd: in an alternating zero/positive run, σ_{i+k} has the opposite status to σ_i. $\sigma_i \geq 0$: "zero or positive" covers all cases.
Thm 1(a)	$k \geq 3$ odd, $n > k$, n odd	Prop. 7, $D \geq 2g$, Lemma 5(iii), Lemma 8	n odd: $k(n+1)/2 \in \mathbb{Z}$, so L is an integer and $\mathsf{msum}(n,k) \in \mathbb{Z}$. k odd: Lemma 8. $n > k$: Lemma 5(i).
Thm 1(b)	$k \geq 3$ odd, $n > k$, n even	Prop. 7, $D \geq 2g$	k odd, n even: $k(n+1)$ odd, so $\mathsf{msum} \in \frac{1}{2} + \mathbb{Z}$ and $L = \frac{k(n+1)+1}{2} \in \mathbb{Z}$; g odd gives $2g \mid n$.
Remark 9	n odd, $0 < k < n$	Lemma 5(i)	n odd: $L = k(n+1)/2 \in \mathbb{Z}$.
Lemma 10	good array	definitions	proved in full in §2 (Claim 1).
Lemma 11	$m = 2h+1 \geq 3$, $k = m-2$	definitions	m odd: $2h \equiv -1$ and $\gcd(h,m) = 1$ (Claim 2).
Lemma 12	good array, $0 \leq j < k$	definitions	column j must exist (Claim 3).
Thm 2	k, m odd, $1 \leq k$, $3 \leq m \leq k+2$	Lemmas 10–12	k, m odd and $m \leq k+2$: $(k-m+2)/2$ is a non-negative integer.
Cor. 3	k, m odd, $k \geq 3$, $m \geq 3$	Thm 1(a), Thm 2, Remark 9, (AFG7)	odd m is either $\leq k+2$ or $\geq k+4$ ($k+3$ is even), so the two cases are exhaustive.
Cor. 4	$k = 3$, $n \geq 4$	Thm 1, Cor. 3, Remark 9, (AFG4), four explicit arrangements	case split on the parity of n and on $3 \mid n$; see §3.

The only place where k odd is genuinely needed in the lower bound is Lemma 8 (and the parity bookkeeping in Theorem 1). Theorem 2 needs k and m odd for the base array and for the number of insertions. In the note the hypothesis of Theorem 2 reads "Let $k \geq 1$ and $m \geq 3$ be odd", meaning that both are odd; the next version states it as "Let k and m be odd integers with $k \geq 1$ and $3 \leq m \leq k+2$", and Corollary 3 likewise, to avoid the reading in which only m is odd.

2. The construction, with every index written out

Theorem 2. *Let k and m be odd integers with $k \geq 1$ and $3 \leq m \leq k+2$, and put $n = mk$. Then there is a permutation π of $[n]$ with $s_i \leq \frac{k(n+1)}{2} + 1$ for all i.*

Arrays and staircase sums. An *array* is a map $A : \mathbb{Z}_m \times \{0, \dots, k-1\} \to [m]$ (rows $q \in \mathbb{Z}_m$, columns c). It is *column-complete* if each column $q \mapsto A(q,c)$ is a bijection $\mathbb{Z}_m \to [m]$. For $0 \leq r \leq k$ put

$$W_A(q,r) = \sum_{c=0}^{r-1} A(q+1,c) + \sum_{c=r}^{k-1} A(q,c),$$

the sum of one entry from each column: columns left of r taken from row $q+1$, the others from row q. Note $W_A(q,k) = W_A(q+1,0)$. Call A *good* if it is column-complete and $W_A(q,r) \le \frac{k(m+1)}{2} + 1$ for all $q \in \mathbb{Z}_m$ and $0 \le r \le k-1$ (then also for $r = k$).

Claim 1 (Lemma 10: blocks, any order)**.** *Let A be good and β any bijection of $\{0,\dots,k-1\}$. Define π on the positions $p = qk + c$ $(0 \le q \le m-1,\ 0 \le c \le k-1)$ of a circle of length $n = mk$ by $\pi_{qk+c} = m\beta(c) + A(q,c)$. Then π is a permutation of $[n]$ and every k-consecutive sum of π is at most $\frac{k(n+1)}{2} + 1$.*

Proof. Permutation. Fix c. As q runs over $\mathbb{Z}_m$, $A(q,c)$ runs over $[m]$, so π_{qk+c} runs over the block $B_c = \{m\beta(c)+1,\dots,m\beta(c)+m\}$. Since β is a bijection onto $\{0,\dots,k-1\}$, the blocks B_c are the k disjoint intervals $\{mb+1,\dots,mb+m\}$, $0 \le b \le k-1$, whose union is $[mk]$. The map $(q,c) \mapsto qk+c$ is a bijection onto $\{0,\dots,n-1\}$, so π is a permutation of $[n]$.

Windows. Every window of k consecutive positions starts at some $p = qk + r$ with $q \in \mathbb{Z}_m$ and $0 \le r \le k-1$. Its positions are $qk+r,\dots,qk+k-1$ (row q, columns $r,\dots,k-1$) followed by $(q+1)k,\dots,(q+1)k+r-1$ (row $q+1$ taken modulo m, columns $0,\dots,r-1$); when $q = m-1$ the second group is row 0, because position mk is position 0. So the window contains exactly one position in each column c, in row $q+1$ if $c < r$ and in row q if $c \ge r$. Its sum is therefore

$$\sum_{c=0}^{k-1} m\beta(c) + W_A(q,r) = m \cdot \frac{k(k-1)}{2} + W_A(q,r),$$

because $\sum_c \beta(c) = 0 + 1 + \cdots + (k-1)$ for *every* bijection β. This is where the block order drops out: it changes which block a column uses, but each window always meets every block exactly once. Finally

$$m\frac{k(k-1)}{2} + \frac{k(m+1)}{2} + 1 = \frac{mk^2 - mk + km + k}{2} + 1 = \frac{k(mk+1)}{2} + 1 = \frac{k(n+1)}{2} + 1. \quad \square$$

Claim 2 (Lemma 11: base array)**.** *Let $m = 2h+1 \ge 3$ and $k = m-2$. Define $C : \mathbb{Z}_m \to [m]$ by $C(0) = m$, $C(y) = m - y$ for $1 \le y \le h$, and $C(y) = y - h$ for $h+1 \le y \le 2h$. Then $A(q,c) = C(q + hc)$ (indices modulo m) is good.*

Proof. C is a bijection. Its values are m; $m-1,\dots,m-h = h+1$; and $1,\dots,h$: together $[m]$, each once. For each c the map $q \mapsto q + hc$ is a bijection of $\mathbb{Z}_m$, so A is column-complete.

Rewriting row $q+1$. Since $2h = m-1$, we have $-2h \equiv 1 \pmod m$, so $q + 1 + hc \equiv q + h(c-2)$ and $A(q+1,c) = C(q + h(c-2))$. Hence, for $0 \le r \le k-1$,

$$W_A(q,r) = \sum_{c=0}^{r-1} C\big(q+h(c-2)\big) + \sum_{c=r}^{k-1} C(q+hc) = \sum_{j \in J} C(q+hj), \qquad J = \{-2,\dots,r-3\} \cup \{r,\dots,k-1\},$$

substituting $j = c-2$ in the first sum and $j = c$ in the second. The two ranges are disjoint, and $J = \{-2,-1,\dots,k-1\} \setminus \{r-2, r-1\}$, where the removed pair always lies inside $\{-2,\dots,k-1\}$ because $0 \le r \le k-1$.

Completing the sum. The set $\{-2,\dots,k-1\}$ consists of $k+2 = m$ consecutive integers. As $\gcd(h,m) = \gcd(h,2h+1) = 1$, the residues $q + hj$ for j in any m consecutive integers are all of $\mathbb{Z}_m$, each once. So $\sum_{j=-2}^{k-1} C(q+hj) = \sum_{y \in \mathbb{Z}_m} C(y) = \frac{m(m+1)}{2}$, and with $y = q + h(r-2)$,

$$W_A(q,r) = \frac{m(m+1)}{2} - C(y) - C(y+h).$$

The key inequality $C(y) + C(y+h) \ge m$ for all $y \in \mathbb{Z}_m$. Three cases.

- $y = 0$: $C(0) + C(h) = m + (m-h) = m + h + 1 > m$.
- $1 \le y \le h$: then $h+1 \le y+h \le 2h$, so $C(y) + C(y+h) = (m-y) + (y+h-h) = m$.

- $h+1 \le y \le 2h$: then $2h+1 \le y+h \le 3h$, so $y+h \equiv y+h-m = y-h-1 \in \{0,\dots,h-1\}$. If $y-h-1=0$ then $C(y+h)=C(0)=m$; otherwise $C(y+h)=m-(y-h-1)$; in both cases $C(y+h)=m-y+h+1$. With $C(y)=y-h$ the sum is $m+1$.

Hence $W_A(q,r) \le \frac{m(m+1)}{2} - m = \frac{m(m-1)}{2}$, and with $k=m-2$, $\frac{k(m+1)}{2}+1 = \frac{(m-2)(m+1)+2}{2} = \frac{m(m-1)}{2}$. So A is good. □

Claim 3 (Lemma 12: adding two columns)**.** *Let A be good with k columns, $0 \le j \le k-1$, and let $U = A(\cdot, j)$ and $\bar{U} = m+1-U$. Let A' have the $k+2$ columns: old columns $0,\dots,j-1$; then U (new index j); then $\bar{U}$ (new index $j+1$); then old columns $j,\dots,k-1$ (new indices $j+2,\dots,k+1$). Then A' is good.*

Proof. U is a column of A and $\bar{U}$ is its image under $x \mapsto m+1-x$, so both are bijections onto $[m]$. For each row, $U(q)+\bar{U}(q) = m+1$. Fix q and $0 \le r' \le k+1$, and recall that $W_{A'}(q,r')$ takes new column c' from row $q+1$ if $c' < r'$ and from row q otherwise.

- $r' \le j$. Both new columns ($c' = j, j+1 \ge r'$) come from row q and give $m+1$. Old column c has new index c if $c < j$ and $c+2$ if $c \ge j$; it comes from row $q+1$ exactly when its new index is $< r'$, i.e. when $c < r'$ (since $r' \le j$). So $W_{A'}(q,r') = W_A(q,r') + (m+1)$.
- $r' \ge j+2$. Both new columns come from row $q+1$ and give $m+1$. Old column $c < j$ (index $c < r'$) comes from row $q+1$; old column $c \ge j$ (index $c+2$) comes from row $q+1$ exactly when $c < r'-2$. As $r'-2 \ge j$, in both cases the rule is "$c < r'-2$", so $W_{A'}(q,r') = W_A(q,r'-2) + (m+1)$, where $0 \le r'-2 \le k-1$.
- $r' = j+1$. U (index $j < r'$) comes from row $q+1$, $\bar{U}$ (index $j+1$) from row q. Old columns $c < j$ come from row $q+1$ and $c \ge j$ from row q. So $W_{A'}(q,j+1) = \sum_{c<j} A(q+1,c) + A(q+1,j) + \big(m+1-A(q,j)\big) + \sum_{c \ge j} A(q,c) = \sum_{c \le j} A(q+1,c) + \sum_{c>j} A(q,c) + (m+1) = W_A(q,j+1) + (m+1)$.

In every case $W_{A'}(q,r') \le \frac{k(m+1)}{2} + 1 + (m+1) = \frac{(k+2)(m+1)}{2} + 1$, which is the bound for $k+2$ columns. □

Proof of Theorem 2. Let $k_0 = m-2$, which is odd and ≥ 1. Claim 2 gives a good $m \times k_0$ array. Since k and m are odd and $m \le k+2$, $t = (k-k_0)/2 = (k-m+2)/2$ is a non-negative integer. Applying Claim 3 t times (with any choice of the insertion column j each time) gives a good $m \times k$ array; Claim 1 with any bijection β gives the permutation. When $m = k+2$, $t = 0$ and the base array is used directly. □

Remark. The proof never restricts the block order β or the insertion positions: every choice works. The programs sample random block orders and several insertion sequences only as a check of the code, not as part of the proof.

3. Boundary cases

- $k = 1$ (allowed in Theorem 2): then $m = 3$, $n = 3$, and the bound $\frac{1\cdot 4}{2} + 1 = 3$ is trivial.
- $m = 3$: $k_0 = 1$ and $C = (3,2,1)$; the base array is one column. For $k = 3$ one insertion gives $n = 9$ with all 3-sums ≤ 16; the arrangement $(9,4,3,8,5,2,7,6,1)$ in the note is of this form, so $a(9) \le 16$, and $a(9) \ge 16$ by Remark 9. This is one of the four exceptions; it is consistent with Theorem 1(a), which needs $9 > (3+3)\cdot 3 = 18$ and does not apply.
- $m = k+2$: no insertion; e.g. $(k,m) = (3,5)$ gives $n = 15$ and $a(15) \le 25$, with equality by Remark 9.
- $n = 21$, $k = 3$: $g = 3$ and $21 > 18 = (k+3)g$, so Theorem 1(a) gives $\mathsf{msum}(21,3) \ge 2$; also $m = 7 = k+4$ in Corollary 3. The upper bound $\mathsf{msum}(21,3) \le 2$ comes from (AFG4) or from Schoenfield's arrangement. So $a(21) = 35$ by a complete argument, independently of the personal communication cited by HK.

- $n = 2k$: excluded in Theorem 1(b), and genuinely so, since $\mathsf{msum}(2k, k) = \frac{1}{2}$ (AFG, Theorem 2). For $k = 3$ this is $n = 6$, one of the four exceptions; there $a(6) \geq \lceil 21/2 \rceil = 11$ trivially and $(6, 1, 4, 5, 2, 3)$ attains 11.
- $n = 4$, $k = 3$: each window omits one number, so every window sum is $10 - \pi_i$ and $a(4) = 10 - 1 = 9 = f(4)$.
- $n = 5$: odd, $3 \nmid 5$, and $5 \leq 6 = (k + 3)g$, so Theorem 1(a) does not apply; $(5, 1, 4, 3, 2)$ attains $10 = f(5) - 1$ and Remark 9 gives $a(5) \geq 10$.
- *Odd* $n \geq 7$ *with* $3 \nmid n$: Theorem 1(a) with $g = 1$ applies as $n > 6$. *Odd multiples of* 3: Theorem 1(a) with $g = 3$ applies for $n > 18$, i.e. $n \geq 21$; the remaining ones, $n = 9$ and 15, are exceptions. *Even* $n \geq 4$, $n \neq 6$: Theorem 1(b).

These cases are exhaustive for $n \geq 4$, so Corollary 4 follows.

4. What the computations do and do not establish

They are evidence that the statements and the code agree, never a step of a proof.

- *Two independent implementations of the lower-bound inequalities.* `tests/lemmas.c` (C) and the function `check` in `stress_lemmas.py` (Python) were written separately from the definitions. The two Python stress scripts share `check`, so they are one implementation tested on different inputs, as the reviewer notes. `lemmas.c` covers every arrangement for $n \leq 11$ ($k = 3, 5$); `check` covers large and near-optimal arrangements.
- *The construction* is checked by computing all window sums of the produced permutation directly; this does not reuse the array formulas of the proof.
- *The exhaustive searches* (`clock_bt3.c`, `kwin.c`) were tested against plain enumeration (`brute.c`) and reproduce all published terms $a(3), \ldots, a(38)$.